

EMG-to-Text Decoding with Recurrent and Attention-Based Architectures

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What We Built

Surface electromyography (sEMG) signals recorded from the wrist can be decoded into typed text, but the mapping from muscle activity to keystrokes is unsegmented: characters are typed at arbitrary timestamps with no ground-truth alignment to the signal. Using the emg2qwerty dataset (32 channels of wrist sEMG paired with keystroke logs from a single subject), we built on a provided Time-Depth Separable (TDS) convolutional baseline and systematically compared recurrent architectures (GRU, LSTM) against attention-based architectures (Transformer, CNN+Transformer, TDSConv+Transformer, Conformer) for character-level decoding under Connectionist Temporal Classification (CTC) loss. The bidirectional GRU was the strongest model in the study, reducing test character error rate (CER) from a 23.77% baseline to 13.98% after 40 epochs.

My Role

My contribution centered on two of the six architecture families the group evaluated: the recurrent GRU configurations and the base attention-based Transformer. For the GRU family, I configured and ran each variant as an individual, controlled case: a unidirectional GRU sized to match the baseline's parameter count for a fair comparison, a bidirectional version to test whether future context improved keystroke resolution, and a hybrid TDSConv+GRU configuration to test whether convolutional feature extraction ahead of the recurrent layer added anything beyond the GRU alone. Running these as separate, isolated cases rather than jointly tuning them is what let the group attribute the performance gain specifically to bidirectionality rather than to some other confound.

On the attention side, I implemented and tested the base Transformer encoder. It performed the worst of any model in the study, and that failure was informative: it is what motivated the group's later hybrid designs, which paired convolutional front-ends with attention specifically because pure self-attention could not capture the local temporal structure in the raw EMG signal on its own.

How We Measured It

For the GRU family, I ran three configurations against the shared TDS convolutional baseline (22.51% validation CER, 23.77% test CER after 40 epochs), holding the rest of the pipeline constant so any performance difference could be attributed to the architectural change being tested:

Configuration	Val CER	Test CER	Parameters
GRU (unidirectional)	29.38%	30.88%	6.0M
GRU (bidirectional)	15.11%	13.98%	14.7M
TDSConv + GRU (unidirectional)	23.11%	22.41%	6.0M

Results after 40 epochs of training, single subject, seed 1501.

The unidirectional GRU, sized to match the baseline's 5.3M parameters for a fair comparison, performed worse than the convolutional baseline outright. Adding bidirectionality reversed that result entirely, cutting test CER by more than half relative to the unidirectional version, at the cost of roughly 2.5× the parameters. The hybrid TDSConv+GRU configuration improved on the plain unidirectional GRU but did not come close to the bidirectional version, indicating that the gain came specifically from bidirectional context rather than from any convolutional front-end. With extended training to 150 epochs, the bidirectional GRU improved further to 13.18% validation CER and 15.83% test CER, the best result of any model or configuration in the study.

For the attention-based family, I implemented and tested the base Transformer encoder: four layers, eight attention heads per layer, sinusoidal positional encodings, and pre-layer normalization for training stability. It reached 57.40% validation CER and 57.45% test CER, more than double the convolutional baseline's error rate, using 16.2M parameters, more than any other model in the study. That result was the strongest evidence in the whole project that self-attention alone does not have the right inductive bias for raw, unsegmented EMG signal, and it is why the rest of the group's attention-based work paired attention with a convolutional front-end rather than using it standalone.

How We Approached the Work

The project used Hydra config files to define each model variant, which meant every architectural change was an explicit, auditable diff against the baseline rather than a change buried in code. The starting point was the shared TDS convolutional baseline:

```
module:
  _target_: emg2qwerty.lightning.TDSConvCTCModule
  in_features: 528
  mlp_features: [384]
  block_channels: [24, 24, 24, 24]
  kernel_width: 32
  decoder:
    _target_: emg2qwerty.decoder.CTCGreedyDecoder
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding: [1800, 200]
```

The bidirectional GRU configuration that performed best replaced the convolutional encoder outright, scaling up to a 512-unit, 3-layer bidirectional recurrent stack:

```

module:
  _target_: emg2qwerty.lightning.GRUCTCModule
  in_features: 528
  mlp_features: [384]
  num_features: 768
  hidden_size: 512
  num_layers: 3
  bidirectional: true
  dropout: 0.2
  decoder:
    _target_: emg2qwerty.decoder.CTCGreedyDecoder

datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding: [1800, 200]

```

The hybrid version kept the convolutional front end from the baseline and added a lightweight, single-layer unidirectional GRU after it, isolating whether convolutional feature extraction ahead of a recurrent layer could match what the larger bidirectional GRU achieved on its own:

```

module:
  _target_: emg2qwerty.lightning.TDSCnvGRUCTCModule
  in_features: 528
  mlp_features: [384]
  num_features: 768
  block_channels: [24, 24, 24, 24]
  kernel_width: 32
  proj_size: 256
  hidden_size: 128
  num_layers: 1
  dropout: 0.2
  decoder:
    _target_: emg2qwerty.decoder.CTCGreedyDecoder

datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding: [1800, 200]

```

It didn't: the hybrid reached 22.41% test CER against the bidirectional GRU's 13.98%, which is what let the group conclude that the gain was coming from bidirectional context specifically, not from recurrence in general. The same discipline applied to the Transformer, evaluated separately from this config family: rather than tuning it extensively to try to close the gap with the recurrent models, I ran it as a clean, unmodified baseline for the attention-based architectures, which is what let the group identify that the problem was the inductive bias of pure self-attention rather than a tuning issue specific to one run.